• Row Echelon Form

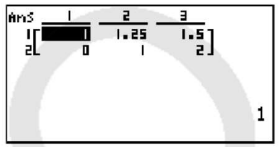
[OPTN]-[MAT]-[Ref]

This command uses the Gaussian elimination algorithm to find the row echelon form of a matrix.

Example **To find the row echelon form of the following matrix:**

Matrix A = $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

[OPTN] [F2](MAT) [F6](▷) [F4](Ref)
[F6](▷) [F1](Mat) [ALPHA] [X,θ,T](A) [EXE]



• Reduced Row Echelon Form

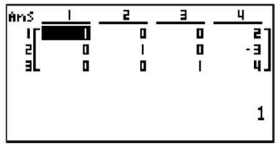
[OPTN]-[MAT]-[Rref]

This command finds the reduced row echelon form of a matrix.

Example **To find the reduced row echelon form of the following matrix:**

Matrix A = $\begin{bmatrix} 2 & -1 & 3 & 19 \\ 1 & 1 & -5 & -21 \\ 0 & 4 & 3 & 0 \end{bmatrix}$

[OPTN] [F2](MAT) [F6](▷) [F5](Rref)
[F6](▷) [F1](Mat) [ALPHA] [X,θ,T](A) [EXE]



- The row echelon form and reduced row echelon form operation may not produce accurate results due to dropped digits.

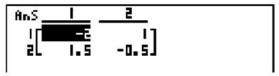
• Matrix Inversion

[x^{-1}]

Example **To invert the following matrix:**

Matrix A = $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

[OPTN] [F2](MAT) [F1](Mat)
[ALPHA] [X,θ,T](A) [SHIFT] [)] (x^{-1}) [EXE]



- Only square matrices (same number of rows and columns) can be inverted. Trying to invert a matrix that is not square produces an error.
- A matrix with a determinant of zero cannot be inverted. Trying to invert a matrix with determinant of zero produces an error.

CASIO